Game Theory in Electricity Markets

Exercise Sheet 1 — Warm-up

4th of September 2026

Exercise 1 Find your own Gijón

In the lecture we saw two examples where rational players followed the rules perfectly, acted in their own best interest, and still produced an outcome that made everyone worse off.

Task. Find one more example of a poorly designed system: a setting in which participants acting rationally within the rules produce an outcome that is collectively undesirable. The example can come from sports, business, politics, everyday life, or elsewhere. You will be surprised how many of these one can find.

Exercise 2 Strategic Games in Normal Form

All players choose their action simultaneously and independently. The cost that player i ends up paying, $c_i(a)$, depends not only on i's own action a_i, but on the entire joint action a — i.e., on what everybody else does too. This is exactly what happened with generator G3 in the lecture: its profit did not depend on its own bid alone, but on its bid *together with* everyone else's bids.

It is equally common to specify a game via **payoff** (or utility) functions $u_i = -c_i$, which player i seeks to *maximize* instead of minimize. The two descriptions are interchangeable; we will use whichever is more natural for the example at hand.

Exercise 3 The Prisoner's Dilemma

Two suspects, players 1 and 2, are arrested and interrogated in separate rooms. Each can either **stay silent** (Cooperate, C) or **betray** the other suspect (Defect, D). The resulting prison sentences (in years) depend on both choices:

- if both stay silent, the police only have enough evidence for a minor charge: **2 years each**;
- if both betray each other: **9 years each**;
- if one betrays and the other stays silent: the betrayer goes free (**0 years**), the one who stayed silent takes the full sentence (**10 years**).

In terms of the definition above:

- **Players:** $N = \{1, 2\}$;
- **Actions:** $A_1 = A_2 = \{C, D\}$;
- **Costs:** $c_i(a)$ = years in prison for player i, given the joint action $a = (a_1, a_2)$, as tabulated below.

		Player 2	
		C	D
Player 1	C	(2, 2)	(10, 0)
	D	(0, 10)	(9, 9)

Each cell shows $(c_1(a), c_2(a))$ *– the cost (years) to Player 1 and Player 2, respectively, under that joint action.*

(Task 3.1) What outcome do you expect two rational, self-interested players to reach? Justify your answer.

(Task 3.2) Comment on whether this game is *well designed*. Can you calculate the price of anarchy?

Exercise 4 Hawk and Dove

Two birds contest a resource of value $V > 0$ (e.g., food, territory). Each animal simultaneously chooses to act aggressively (**Hawk**, H) or passively (**Dove**, D).

- If both play *Dove*, they share the resource peacefully: each gets $V/2$.
- If one plays *Hawk* and the other *Dove*: the Hawk takes almost the whole resource $0.9V$, the Dove retreats and gets only $0.1V$.
- If both play *Hawk*, they fight over it, until they realize they are equally matched, share and both get half of the resource. The catch: fighting comes not for free, it cost both of them C. So in total they both get a payoff $V/2 - C$.

We will use **payoffs** here (higher is better) rather than costs, since that is the natural convention for this game; recall $u_i = -c_i$.

(Task 4.1) Construct the payoff table for this game, in terms of V and C. Use the same format as the Prisoner's Dilemma table above.

		Player 2	
		H	D
Player 1	H	(_____, _____)	(_____, _____)
	D	(_____, _____)	(_____, _____)

(Task 4.2) Assume fights are costly, i.e. $C > V$. What outcome would you expect rational animals to reach? Does your answer change if C becomes smaller? If so, when?

Exercise 5 Rock, Paper, Scissors

Two players simultaneously choose one of three actions: **Rock**, **Paper**, or **Scissors**. Rock beats Scissors, Scissors beats Paper, and Paper beats Rock; identical actions are a tie. The winner receives a payoff of $+1$, the loser -1, and both players get 0 on a tie.

(Task 5.1) Construct the payoff table for this game.

		Player 2		
		Rock	Paper	Scissors
Player 1	Rock	_____	_____	_____
	Paper	_____	_____	_____
	Scissors	_____	_____	_____

(Task 5.2) What outcome would you expect rational players to reach?

Exercise 6 My New Apartment

I recently moved into a new apartment. My partner is currently traveling but on the 1st of October they will also move in. I have almost all furniture except that in my old apartment the bed was owned by the landlord such that I could not bring it. We therefore just decided to bring my partners bed, which is currently in the basement of a friend, somewhere in Jutland. Therefore I currently sleep on an air mattress which is not very comfortable and my back hurts every morning. Every day I can now decide if i want to rent a car, go to Jutland and get the heavy bed or not. If I do not get the bed before the 1st of October my partner will bring it when they move all their other stuff.

With what we have learned so far, help me make a decision.

Hint: assign numbers to how annoying the back pain and getting the bed is.

Tell me precisely if I should get the bed and if so, when. Play around with how you value these two actions. When does your recommendation change? Can you choose evaluations such that your recommendation becomes to get the bed on the 15th of September?

Exercise 7 War of Attrition

Two players are involved in a dispute over an object. The value of the object to player i is $v_i > 0$. Assume $v_1 \neq v_2$. Time is modeled as a continuous variable that starts at 0 and runs indefinitely. Each player chooses when to concede the object to the other player; if the first player to concede does so at time t, the other player obtains the object at that time. If both players concede simultaneously, the object is split equally between them, player i receiving a payoff of $v_i/2$. Time is valuable: until the first concession each player loses one unit of payoff per unit of time.

(Task 7.1) With what we saw in the last exercise, what do you expect the outcome of this game to look like?

(Task 7.2) Formulate this situation as a strategic game and show that the outcome is what you expected it to be.

(Task 7.3) How does it look like if $v_1 = v_2$?

Exercise 8 For next week

Read up on the following topics:

- Nash equilibrium (NE): what is a NE, do they always exist, are they always unique, computationally
- strictly dominated actions: what are they, what is there relation to NE
- mixed NE: what are they, do they always exist, if there are many how does the set look like, how to find them

We have many examples for Nash equilibria and dominated actions in the exercises before. Find them. There is one exercise that does not have either of those two, in this exercise try to find the mixed Nash equilibrium.